Laboratory Work No. 3

Sensorless Control Based on an Adaptive Observer

Variant 3

Goal: to study an adaptive observer with a gradient estimator and use it to build a sensorless PMSM controller.

1. Mathematical Model and Observer

The observer uses model re-parameterization, low-pass filtering, a gradient estimator for the unknown constant vector eta, and a PLL-based speed estimator. The closed-loop sensorless controller replaces the measured rotor position with the observer estimate.

Parameter	Value
Variant	3
L, H	0.033000
R, Ohm	5.3000
Jm, kg*m^2	0.000600
Pole pairs np	5
lambda_m, Wb	0.1600
B, N*m*s/rad	0.0000

Quantity	Value
Kp_i	18.000000
Ki_i	1400.000000
Kp_omega	0.360000
Ki_omega	18.000000
alpha	250.000000
gamma	2.500000
PLL Kp	252.000000
PLL Ki	32400.000000

2. Simulation Scheme

The open-loop observer runs in parallel with the PMSM and FOC. In the final sensorless mode the estimated position is fed back into the FOC block.

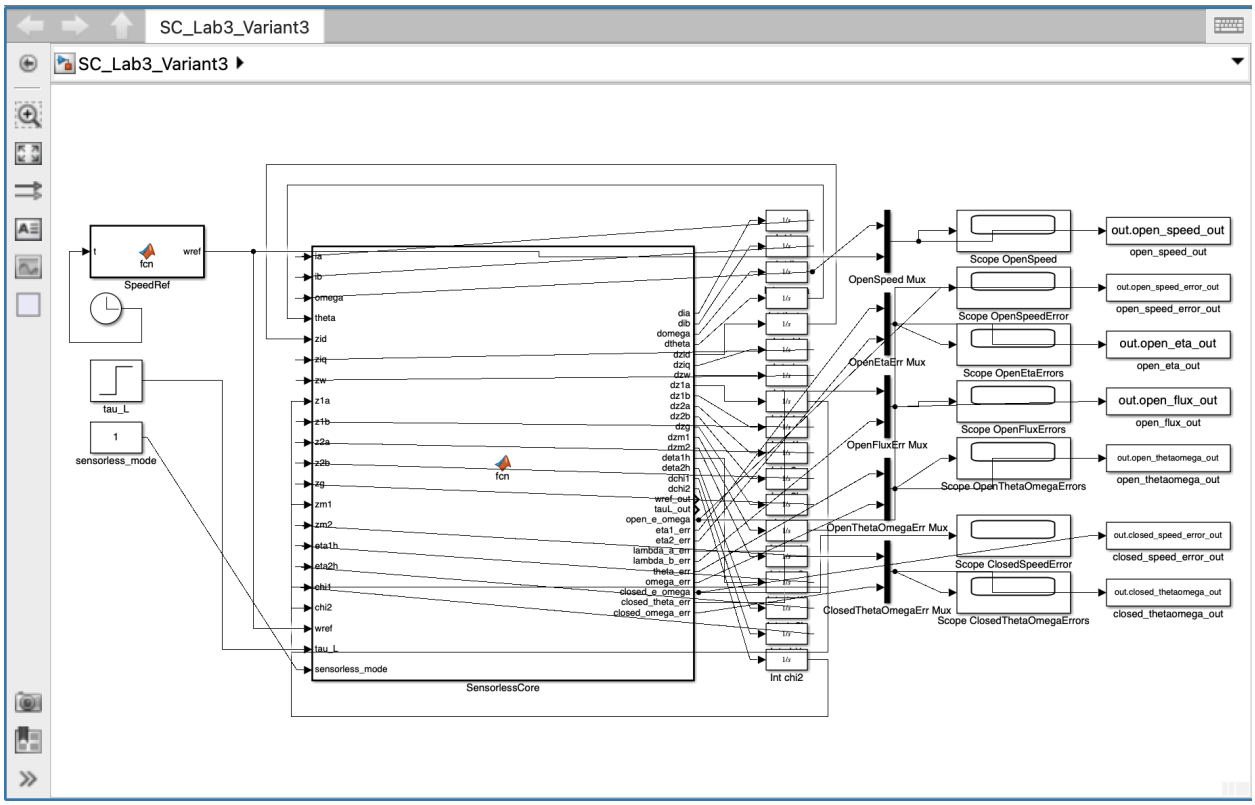

Figure 1. Main Simulink scheme of the adaptive observer and sensorless control system.

3. Simulink Screenshots and Transient Responses

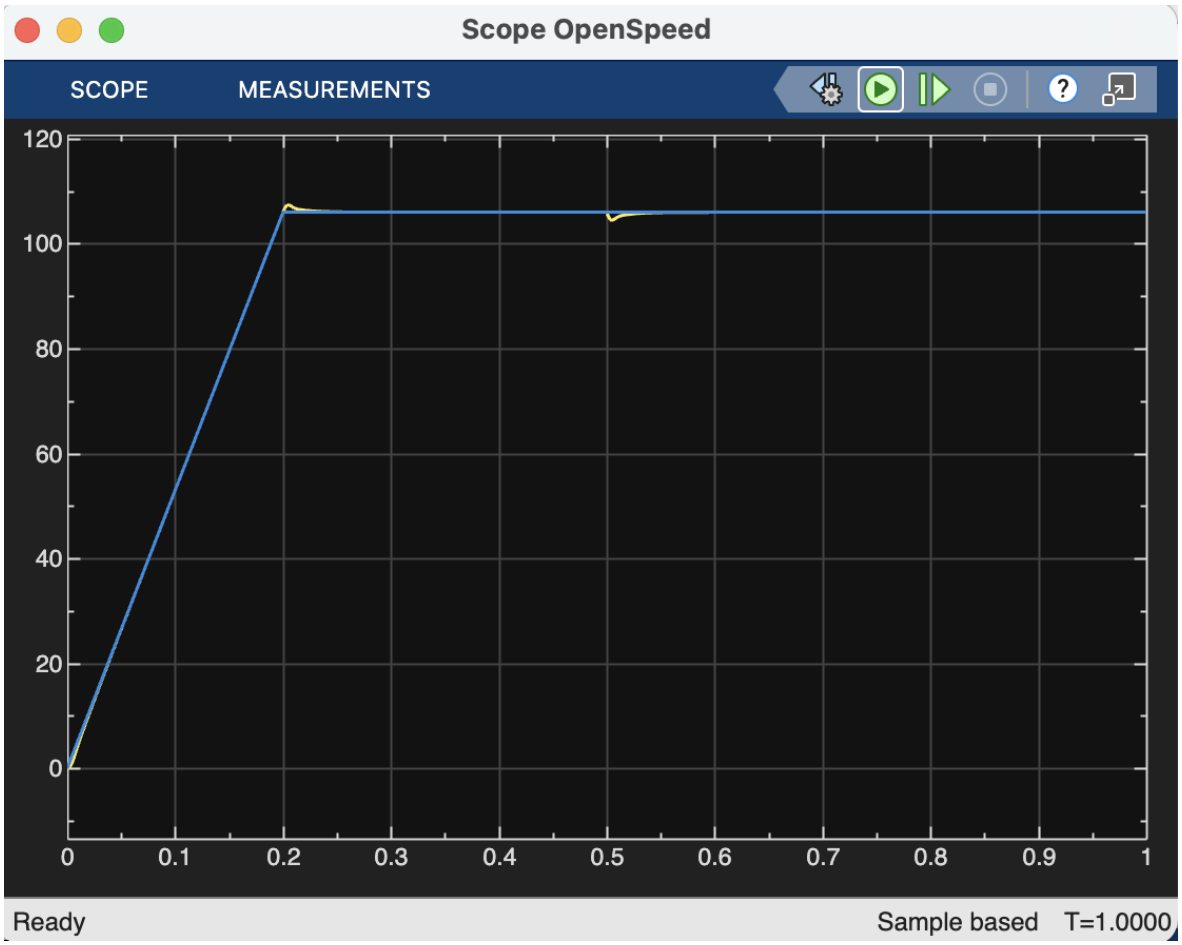

Figure 2. Open-loop observer: measured speed and reference speed.

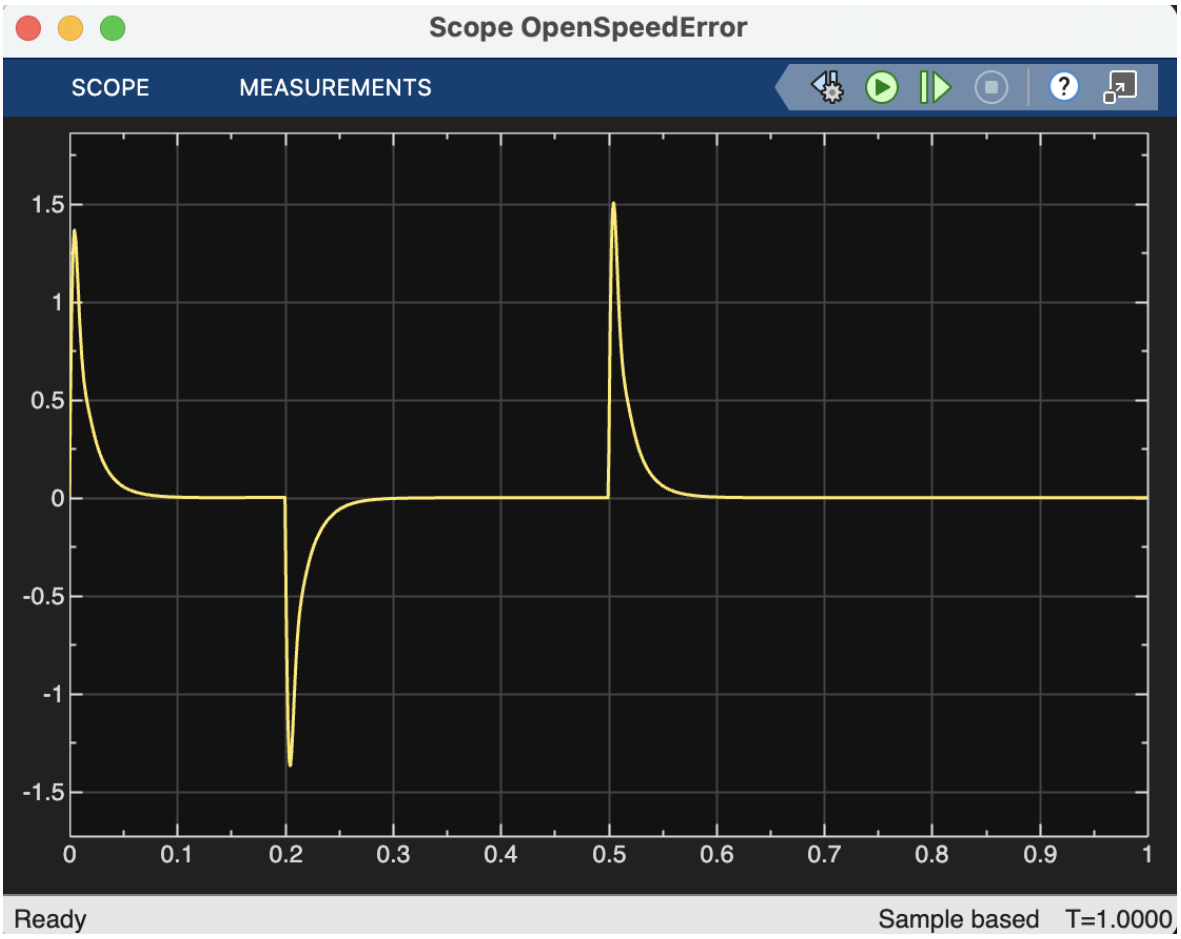

Figure 3. Open-loop observer: speed tracking error.

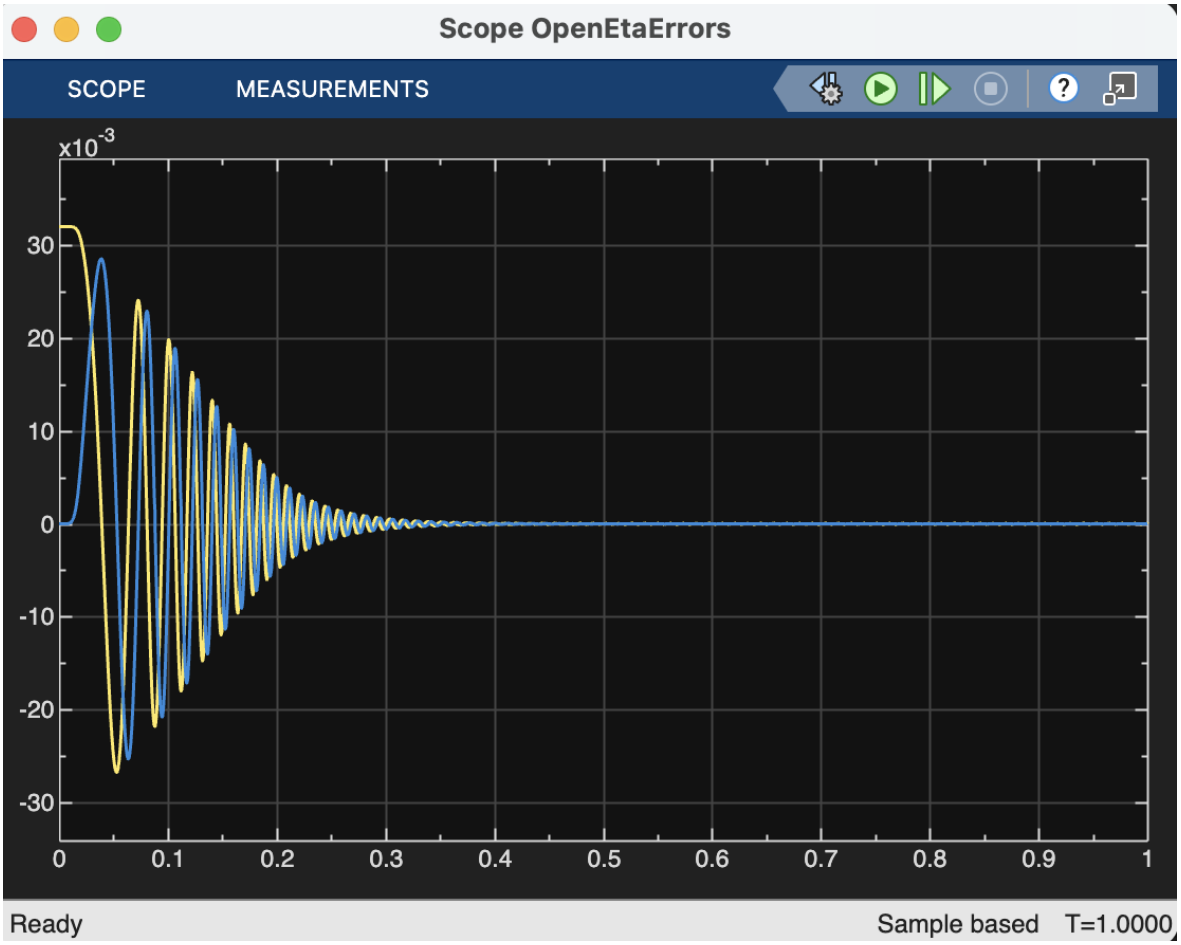

Figure 4. Open-loop observer: parameter estimation errors.

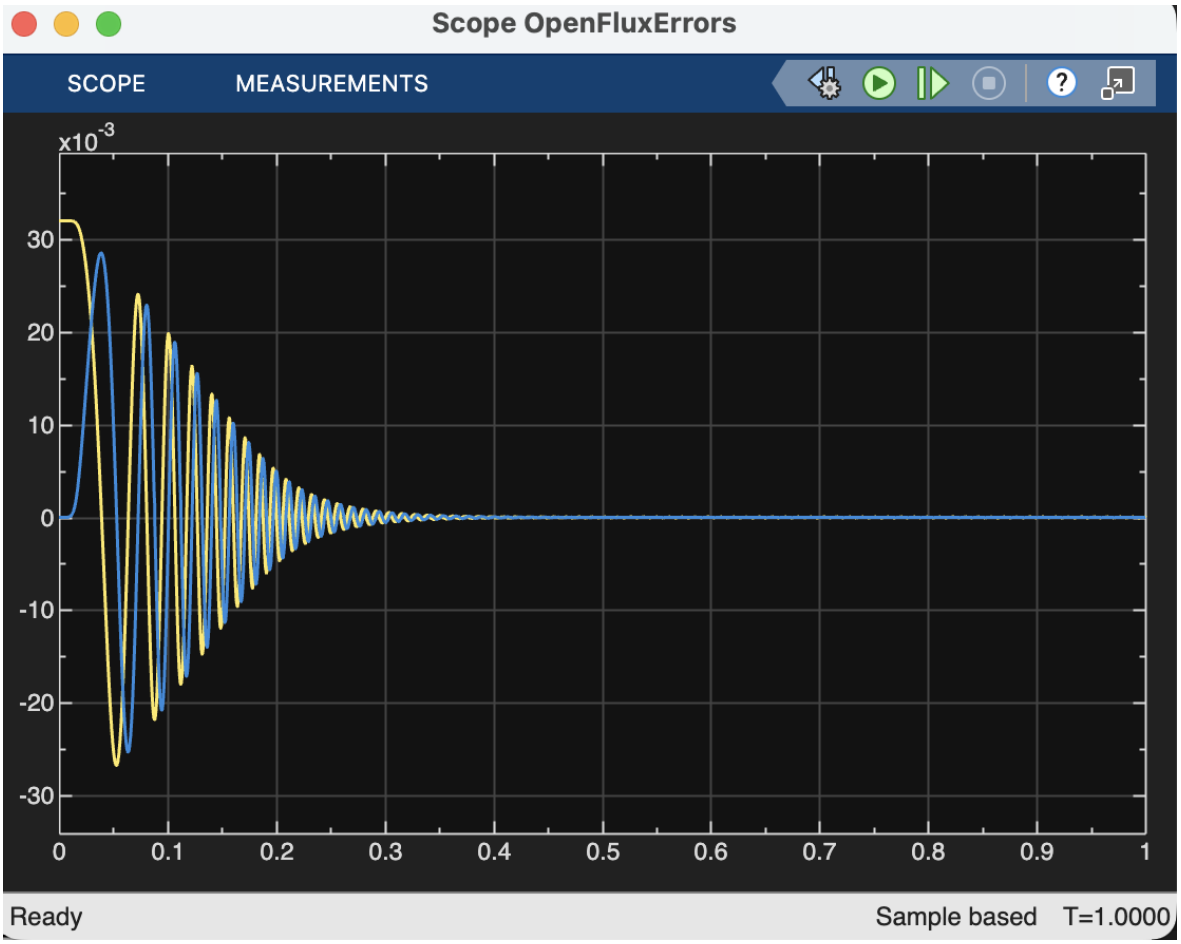

Figure 5. Open-loop observer: total flux estimation errors.

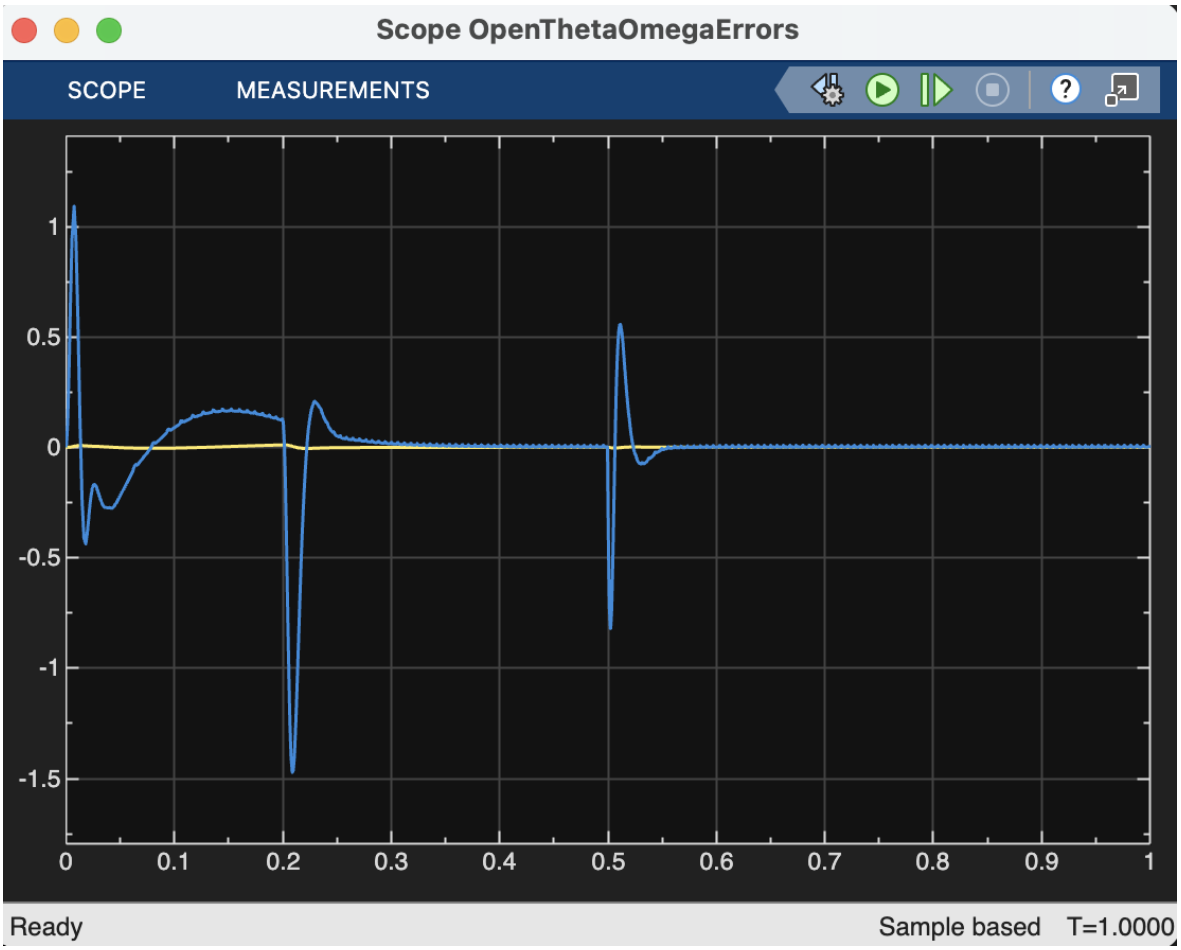

Figure 6. Open-loop observer: position and speed estimation errors.

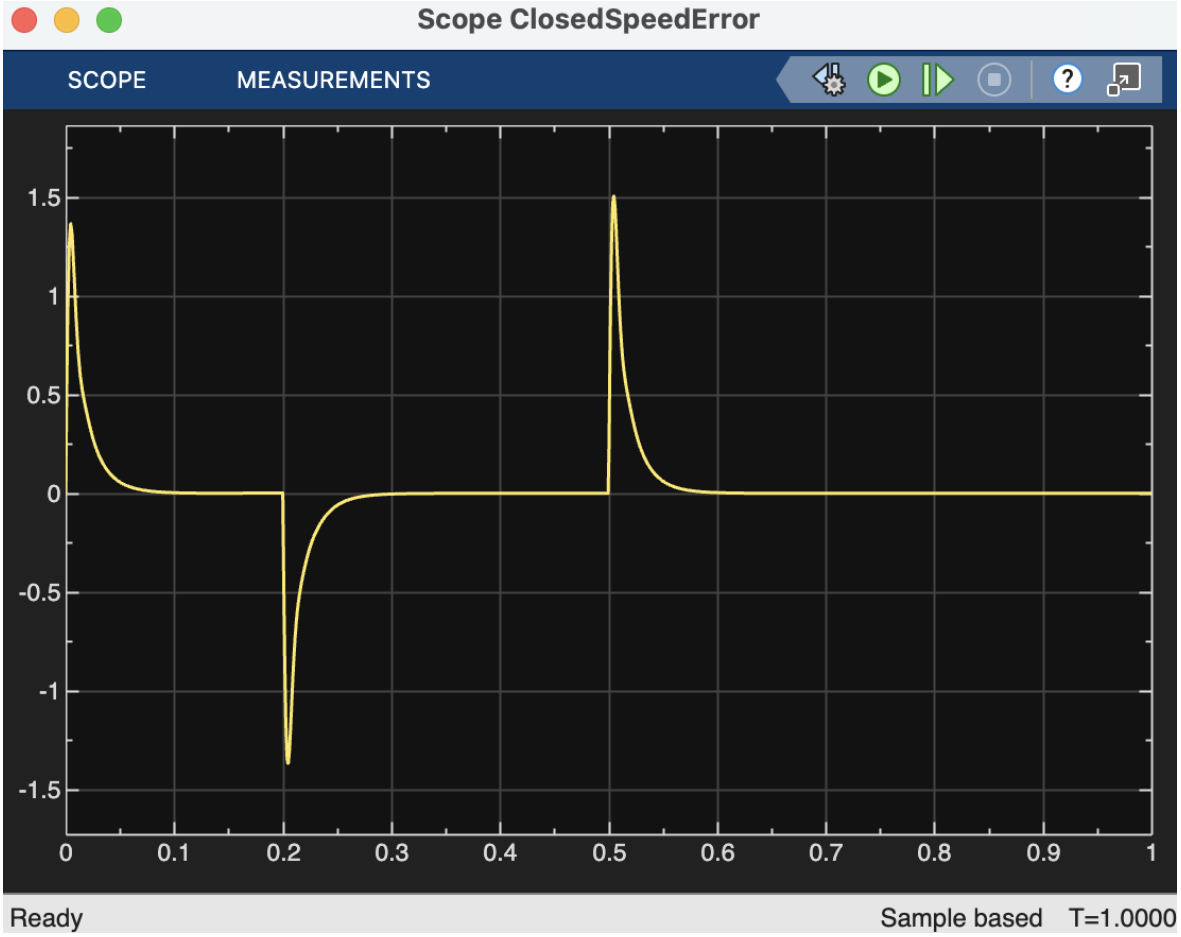

Figure 7. Sensorless controller: speed tracking error.

Figure 8. Sensorless controller: position and speed estimation errors.

5. Selected Values

Quantity	Value
e_omega(0.4), rad/s	-0.000019
e_omega(1), rad/s	0.000000
theta_tilde(0.4), rad	-0.000076
theta_tilde(1), rad	-0.000000
omega_tilde(0.4), rad/s	0.001645
omega_tilde(1), rad/s	0.000001

6. Analysis and Conclusion

The adaptive observer converges to low parameter, flux, position, and speed estimation errors in the steady-state region. After replacing the measured position with the estimated one, the sensorless loop preserves stable speed tracking. This confirms that the observer is sufficiently accurate for closed-loop use under the chosen tuning.